Jiezheng Wei

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(Updated: October 6, 2026)

Education

Ph.D., Economics, Singapore Management University 2022–present
Thesis: AI and High-Frequency Analysis for Monetary Policy Shocks.
Committee: Jia Li (Chair), Andrew J. Patton, Yichong Zhang, Qiyuan Li.
M.Sc., Economics, The Chinese University of Hong Kong, Shenzhen 2019–2021
B.Sc., Economics, Central University of Finance and Economics 2013–2017

Research Interests

Econometrics, AI for Economics.

Working Papers

1. **Job Market Paper:** "Weak Identification in News-jump Regressions."

   Abstract. News-jump regressions are high-frequency event-study regressions under infill asymptotics. When news-jumps in variables of interest over narrow event windows are of the same magnitude as diffusive background noise, the regression coefficient is weakly identified, invalidating estimation and inference of least-squares regressions. We develop identification-robust inference based on Anderson–Rubin tests and derive, in a Gaussian limit experiment arising from the jump-diffusion process, a minimal sufficient and complete statistic for jumps, thereby enabling conditional inference. In applications to FOMC announcements, we invert our identification-robust tests to study the Fed information effect proposed by Nakamura and Steinsson (2018) at higher frequencies and to examine market-factor dependence in news-driven ETF jumps.

2. "Evaluating AI Forecasts on Future Sport Events," with Jia Li and Andrew J. Patton.

   Abstract. We construct an ex-ante, timestamp-verified panel of one-day-ahead forecasts using frontier LLMs on sports events, which allows us to completely avoid lookahead bias. This real-time dataset enables us to answer following questions: (i) Do LLM forecasters outperform the betting markets benchmark? (ii) Are LLM forecasters rational? (iii) Are more

advanced (expensive) LLMs better forecasters? We accordingly answer these questions by evaluating the LLM forecasts on proper score functions, calibration, and panel analysis.

3. "Identifying Monetary Policy Shock with Large Language Models," with Sen Zhang.

   Abstract. We use large language models to measure sentiment indicators from pre-FOMC briefing materials and form Federal Reserve staff aggregate expectations about the US economy. The resulting low-dimensional AI-based measurement captures economically meaningful information by letting AI read semantic context while avoiding regularization bias caused by high dimensionality. As a complement to the existing narrative approaches, our AI-based monetary policy shocks provide a reasonable alternative for downstream impulse response analysis.

4. "A Fixed-k Bipower Jump Test," with Jia Li and Qiyuan Li.

   Abstract. We develop a jump test for Itô semimartingales observed at high frequency based on the ratio of untruncated realized variance to bipower variation within a block of k consecutive returns, where k is held fixed. Under infill asymptotics, the limiting distribution of the proposed test is asymptotically pivotal under the null of no jumps, according to a coupling result inherited from the driving Brownian diffusion, and therefore delivers favorable statistical properties over short time windows. Monte Carlo experiments calibrated to empirical financial data show that the proposed test attains essentially nominal size for small k. The framework provides a single, tuning-free, block-localized price jump detector that plugs directly into the broader apparatus of fixed-k volatility inference in Bollerslev, Li, and Liao (2021) and the bipower jump test of Barndorff-Nielsen and Shephard (2006).

5. "An Honest Inference for High-Dimensional Linear Models," with Dennis Lim, Jonathan H W Tan and Pengjin Min.

   Abstract. Inference in high-dimensional settings poses significant challenges. Building on the influential work of desparsified LASSO in van de Geer, Buhlmann, Ritov, and Dezeure (2014), we develop an inference procedure that extends their approach in two key directions: (i) we allow inference on linear combinations of the true parameter vector at moderate, potentially diverging rates; and (ii) we accommodate general heteroskedasticity in the data. Under mild regularity conditions, we demonstrate both theoretically and through simulations that our proposed test achieves asymptotically exact size control.

Presentations

2023-26 SMU Econometrics Workshop.

2024 SMU Applied Micro Reading Group and SMU Students Econometrics Workshop.

2021 International Conference on Smart Finance, CUHK-Shenzhen.

Research Experience

Research assistant for Prof. Jia Li, Singapore Management University, 2024 - 2026.

Teaching Experience

Singapore Management University
- TA, Time Series Econometrics (PG), 2026
- TA, Econometric Analysis (PG), 2025
- TA, Intermediate Econometrics (UG), 2025
- TA, Panel Data Analysis (UG), 2023
- TA, Real Estate Economics (UG), 2023, 2025

The Chinese University of Hong Kong, Shenzhen
- TA, Applied Probability and Stochastic Processes in Business (UG), 2022
- TA, Machine Learning in Business (UG), 2021, 2022
- TA, Basic Statistics (UG), 2021
- TA, Advanced Econometrics (PG), 2020

Awards & Honors

Singapore Management University - China Scholarship Council Joint Fellowship, 2022 - 2026.
Presidential Award for Outstanding Master's Students, CUHK-Shenzhen, 2021.
Dean's List, School of Management and Economics, CUHK-Shenzhen, 2021.
First Class Scholarship for Academic Excellence and Dean's List, CUHK-Shenzhen, 2021.

Work Experience

2021-22 Research Associate, SME, The Chinese University of Hong Kong, Shenzhen.
2017-19 Associate, Beijing Dusheng Investment Fund (LP), Beijing, China.

Other Services

2024 - 25 Executive Committee, Graduate Research Student Society, SMU.

Others

AI skills: Claude Code, ChatGPT/Codex, AI API.
Programming skills: Python, R, Matlab, LaTeX.
Languages: English (fluent), Chinese (native).
Hobbies: fitness, badminton, classic music, cook.